

LIÈGE université
Sciences Appliquées

Introduction to stability and control problems

ELEC0447 - Analysis of electric power and energy systems

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Overview

1. The concept of stability

2. A quick history of blackouts

3. Assessing Power System Security

Why control power systems?

- Technical requirements: power system devices are designed so as to operate within well-defined "tolerance regions"
  - around nominal values of voltage V_n: 1 ± 0.1 pu in Europe
  - around nominal value of frequency f_n: 50 ± 0.2 Hz in Europe (in steady state)
  - within the *P*-*Q* capabilities of devices
  - under the current limits of lines and transformers
- Large/persistent deviations from nominal values could lead to
  - damages and safety problems (e.g. high voltage)
  - cascading phenomena
  - service interruptions

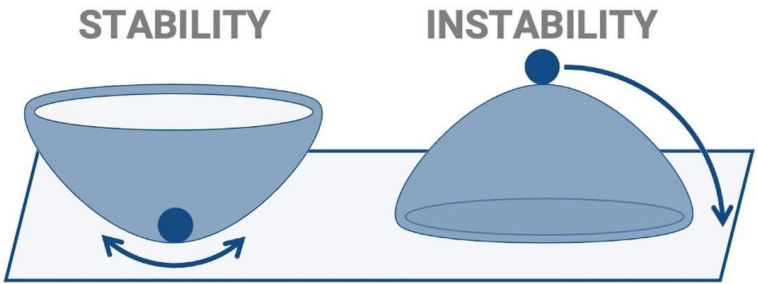
Exogenous threats

- Sudden disturbances, such as line or generator tripping
- Fast variations of the *net load* (cf. Duck curve)
  - the net load refers to the load "seen" by the transmission system, i.e. the load minus the non-controllable dispersed generation
- weather conditions, such as storms, which can impact the generation of renewable energy sources (RES) (e.g. wind turbines' cut-out speed)

1. The concept of stability

General concept

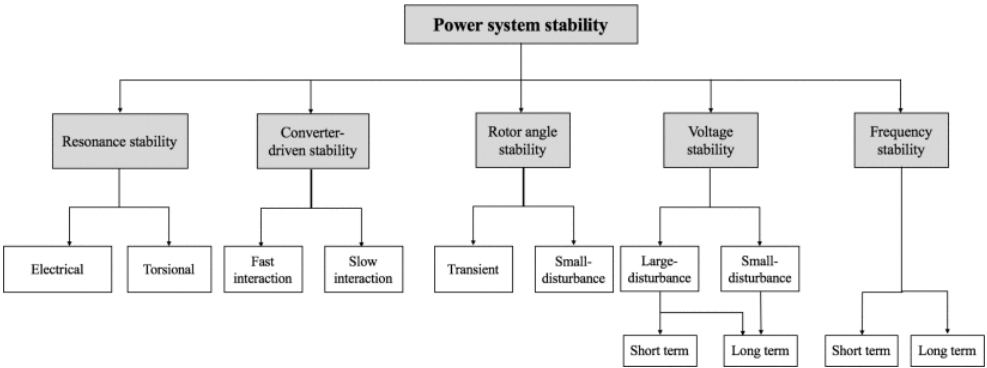
If a system has the property that it will get back into the equilibrium state again after moving away from its equilibrium state, then it is stable. *[1]*



In power systems I

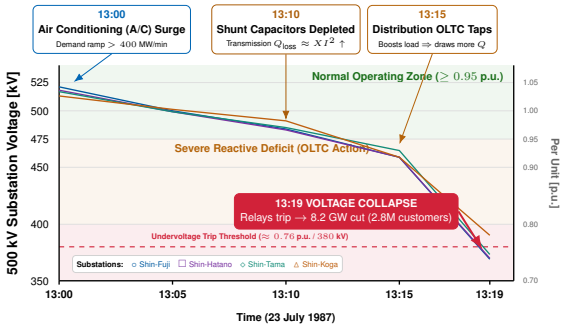
> ***Power system stability is the ability of an electric power system, for a given initial operating condition, to regain a state of operating equilibrium after being subjected to a physical disturbance, with most system variables bounded so that practically the entire system remains intact.*** *[2]*

In power systems II



2. A quick history of blackouts

Tokyo Blackout (23 July 1987)



Chronology of Voltage Collapse [3, 4]:

13:00 — **Air Conditioning (A/C) Surge:**
Heatwave ($> \ 39^{\circ}$C); A/C load surges ($>$ 400 MW/min, $+30\%$ vs forecast).

13:10 — **Capacitors Exhausted:**
Heavy transfers cause line reactive losses $Q_{\text{loss}} \approx XI^2$; shunt capacitors depleted.

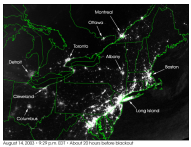
13:15 — **OLTC Tap Restoration:**
Distribution transformers boost voltage to sustain load, drawing even more Q from transmission.

13:19 — **Runaway Voltage Collapse:**
$V < 0.75$ p.u. Undervoltage relays trip lines: **8.2 GW cut**, **2.8M customers** lost.

Pure Voltage Instability

No short-circuit fault occurred. Collapse was purely driven by load dynamics, line reactive losses, and tap-changer response!

Northeast USA & Canada Blackout (14 August 2003)



August 14, 2003 – 4:29 pm EDT • About 20 hours before blackout

August 15, 2003 – 9:12 pm EDT • About 7 hours after blackout

Impact [5]

61.8 GW lost, **50M people** impacted across 8 US states & Ontario. Estimated damage: **$4–10 Billion**.

Chronology of Cascading Failure [5, 6]:

13:31 — **Eastlake Unit 5 Trips (Ohio):**
Major coal plant near Cleveland trips, removing critical local reactive voltage support.

14:14 — **EMS Alarm Processor Freezes (Software Bug):**
Race condition in FirstEnergy's SCADA/EMS alarm logger. Operators fly blind for $>$ 1 h.

15:05 — **345 kV Lines Sag into Trees:**
Heavily overloaded Stuart-Atlanta & Harding-Chamberlin lines overheat and sag into unpruned trees.

16:10 — **Cascading Grid Collapse (7 Minutes):**
Overloads cascade into NY, Ontario & Michigan. **263 power plants trip in 7 minutes**.

Italian Blackout (28 September 2003)

Chronology: Intertie Trips to Island Collapse [7, 6]:

03:01 **Lukmanier 380 kV Line Trips (Switzerland):**
Italy importing 6.7 GW (near limit). Line sags into tree and flashes over.

03:11 **Coordination Delay (Voice Protocol):**
Swiss TSO calls Italy to curtail imports; phone protocol takes 10 min, too slow to relieve lines.

03:25 **San Bernardino Trips → Intertie Cascade:**
Line trips on overload; within 12 seconds, all Swiss, French, and Austrian interties trip!

03:26 **Island Frequency Collapse:**
Instantaneous 6.4 GW deficit (25% demand); f crashes to 48.2 Hz in seconds → blackout.

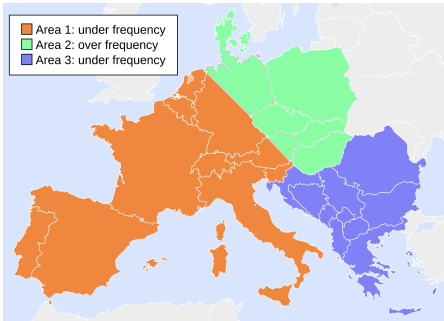
Active Power Imbalance

Interconnected grids depend strictly on continuous active power balance. An unhedged deficit causes immediate frequency collapse!

Impact

27 GW lost, **56M people** affected across Italy (blackout lasted up to 18 hours).

European Grid Disturbance (4 November 2006): The 3 Islands



The Continental Split [8, 9]:

- **Context:** High wind generation in Northern Germany; heavy north-to-south and east-to-west transit flows.
- Cascading line trips split Continental Europe into **three asynchronous islands** in 15 seconds:
  - **Western Island:** Deficit of 8.9 GW (f dropped to 49.0 Hz). Saved by 17 GW automatic UFLS load shedding.
  - **North-Eastern Island:** Surplus of 10.9 GW (f climbed to 51.4 Hz). Stabilized by wind generator overfrequency trips.
  - **South-Eastern Island:** Slight surplus ($f = 50.3$ Hz).

European Disturbance: Timeline of the Continental Split

Chronology of the Split and Restoration [8]:

21:38 — **Planned Line Outage Moved Forward:**
Conneforde-Diele 380 kV line de-energized 2 hours early for ship passage without re-running $N - 1$ check or warning neighbors.

22:05 — **Critical Corridor Overloading:**
Landesbergen-Wehrendorf 380 kV line exceeds thermal limits; manual bus coupling attempt worsens power flow.

22:10 — **Cascade Splits Continent (15 Seconds):**
Line trips on overload; cascade fractures Continental Europe into 3 islands. UFLS sheds 17 GW in Western island.

22:45 — **Resynchronization & Restoration:**
Gradual resynchronization of the 3 islands. 15M European consumers disconnected.

Legacy for Europe

Showed that local operational errors propagate cross-border in seconds. Led directly to:

- Creation of **ENTSO-E** (2009)
- Legally binding **EU Network Codes**
- Regional Security Coordinators (RSC)

Brazilian National Blackout (15 August 2023)

Chronology: Line Trip to Inverter-Driven Split [10]:

08:31:00 — **Quixadá-Fortaleza 500 kV Line Trips:**
High wind/solar export from Northeast to Southeast. Line trips on protection malfunction.

08:31:01 — **IBR Fast Voltage Control Failure:**
Wind/solar plant inverters fail to inject fast reactive current during transient; severe voltage dips propagate.

08:31:04 — **Loss-of-Synchronism Tripping:**
Voltage dip wave triggers loss-of-synchronism and distance protections across 500 kV corridors.

08:31:10 — **National Split & Load Shedding:**
System splits into islands; UFLS disconnects **18.9 GW** ($\sim$ 29% of national demand, 29M people).

Modern Challenge: IBR

Displacing synchronous machines with inverters removes physical inertia and alters voltage dynamics during faults!

Impact

Affected **25 Brazilian states** and the Federal District, lasting up to 6 hours.

Iberian Peninsula Blackout (28 April 2025)

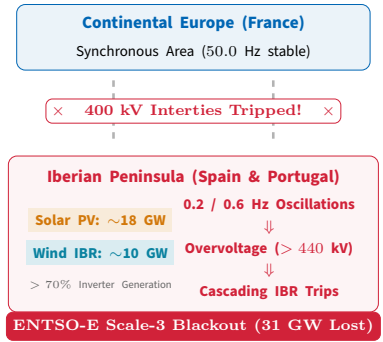
Chronology: Oscillations to Inverter Overvoltage Cascade [11]:

12:03 **Inter-Area Power Oscillations ($>$ 70% IBR):**
Spring midday. Poorly damped 0.21 Hz inter-area and 0.63 Hz local swings develop between Spain & France.

12:25 **TSO Remedial Control Action:**
Operators re-mesh 400 kV lines and set INELFE HVDC to constant power mode to damp oscillations.

12:30 **Fatal Overvoltage Surge ($>$ 440 kV):**
Damping succeeds, but sudden impedance changes trigger extreme voltage surge in reactive-starved grid.

12:32 **Cascading Inverter Trips & Blackout:**
Inverters trip on overvoltage ($>$ 15 GW lost); interties trip on loss-of-synchronism $\rightarrow$ **31 GW lost (55M people)**.



Iberian Blackout: Technical Analysis & Lessons

Root Causes & Inverter Dynamics [11]:

- **Control Coupling:** Shifting HVDC mode and re-meshing lines damped oscillations but triggered the overvoltage surge.
- **Passive Mode:** Solar/wind plants lacked fast dynamic reactive absorption.
- **Nuisance Trips:** Inverters tripped on overvoltage rather than riding through.

Recommendations for High-Renewable Grids:

- **Grid-Forming (GFM):** Converters must supply virtual inertia and voltage support.
- **HVDC POD Tuning:** Coordinated damping without adverse voltage coupling.
- **TSO-DSO Coordination:** Unified reactive management to prevent overvoltages.

New Stability Paradigm

Europe's first major blackout driven by **converter-driven stability** and **fast voltage dynamics**, proving that low-inertia grids require active grid-forming controls!

3. Assessing Power System Security

Operational Security: The $N - 1$ Criterion

How do Transmission System Operators (TSOs) prevent blackouts?

- **Basic principle:** The grid must withstand the loss of any single unexpected critical element ($N - 1$) without causing cascading outages, voltage collapse, or service interruptions.
- **Contingency List ($N - 1$):**
  - Loss of any single transmission line or cable
  - Loss of any substation transformer
  - Loss of the largest generating unit or HVDC link
- If a single failure causes another element to overload beyond emergency ratings, the operating state is **insecure**.

TSO Security Scope

TSOs evaluate both:

1. **Static Security:** Post-fault power flow within thermal and voltage bounds.
2. **Dynamic Security:** Transient rotor angle, frequency, and voltage stability.

Static $N-1$ Security Assessment Workflow I

Real-Time Online Security Analysis (every 5 to 15 minutes in EMS):

1. **State Estimation Base Case (N):**
   - Obtain real-time bus voltages $\mathbf{V}$, injections $\mathbf{P}$, $\mathbf{Q}$, and branch flows from SCADA and PMU measurements.
   - Verify that pre-contingency variables satisfy continuous operating limits:

$$I_{ij}^{(0)} \leq I_{ij}^{\max,\text{perm}}, \quad V_k^{\min,\text{perm}} \leq V_k^{(0)} \leq V_k^{\max,\text{perm}}$$

2. **Simulate Contingency List $\{c = 1, \ldots, C\}$:**
   - For each contingency c, solve the post-outage AC power flow equations:

$$\mathbf{P}_{\text{inj}}^{(c)} = \mathbf{P}(\mathbf{V}^{(c)}, \boldsymbol{\theta}^{(c)}), \quad \mathbf{Q}_{\text{inj}}^{(c)} = \mathbf{Q}(\mathbf{V}^{(c)}, \boldsymbol{\theta}^{(c)})$$

Static $N-1$ Security Assessment Workflow II

3. **Check Post-Contingency Bounds:**
   - Branch currents must respect short-term emergency thermal ratings ($I_{ij}^{(c)} \leq I_{ij}^{\max,\text{emerg}}$).
   - Substation voltages must remain within emergency limits (typically $[0.90, 1.10]$ p.u.).
4. **Industrial Complexity (e.g., Elia in Belgium):**
   - $1000-2000$ single contingencies evaluated.
   - Over 4000 branch and voltage constraints checked per contingency.
   - If violations occur, the operator must enact **preventive redispatch** or prepare fast **corrective remedial actions**.

From Static Security to Dynamic Stability

Static $N-1$ analysis assumes the system reaches a steady state!

- But will the system survive the dynamic transition after a fault?
- **ELEC0447 Core Curriculum:**
  1. **Frequency Control:** Balancing active power ($P-f$) across milliseconds to hours.
  2. **Voltage Stability:** Reactive power balance ($Q-V$) and PV/QV curve margins.
  3. **Rotor Angle Stability:** Machine swing equations, critical clearing time, and synchronism.
  4. **HVDC Integration:** Fast active/reactive power control and converter dynamics.

Next Lecture

Frequency Control: How synchronous inertia, governors, and AGC maintain 50 Hz across the European continent.

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